Chapter 1
SET THEORY
[Part 1: Set & Subset]

Learning outcomes

After this lesson, you can:

Describe a set using a list or a rule.

Distinguish membership, subset and proper subset.

Compare equal, equivalent, finite and infinite sets.

Determine cardinality and list a power set.

Introduction

Why are we studying sets

- The concept of set is basic to all of mathematics and mathematical applications.
- Serves as a basis of description of higher concept and mathematical reasoning
- Set is fundamental in many areas of Computer Science.

Set

- A set is a **well-defined collection of distinct objects.**
- These objects are called **members** or **elements** of the set.
- Well-defined means that we can tell for certain whether an object is a member of the collection or not.
- If a set is finite and not too large, we can describe it by listing the elements in it.

Example

Write the elements inside braces or curly brackets { }.

- *A* is a set of all positive integers less than 10,
  $A=\{1, 2, 3, 4, 5, 6, 7, 8, 9\}$
- *B* is a set of first 5 positive odd integers,
  $B=\{1, 3, 5, 7, 9\}$
- *C* is a set of vowels, $C=\{a, e, i, o, u\}$

Defining Sets

This can be done by:

- Listing ALL elements of the set within braces /curly brackets.
- Listing enough elements to show the pattern then an ellipsis.
- Use set builder notation to define “rules” for determining membership in the set

Three ways to describe a set

Method	Example	What it tells us
Complete list	{2,4,6,8}	Listing ALL elements of the set within braces /curly brackets.
Pattern and ellipsis	{2,4,…,8}	Listing enough elements to show the pattern then an ellipsis. The pattern ends at 8.
Set-builder notation	$\{x \in \mathbb{Z} \mid 2 \leq x \leq 8, x \text{ is even}\}$	A rule decides membership.

Sets

- If a set is a large finite set or an infinite set, we can describe it by **listing a property necessary for memberships**

- Let S be a set, the notation,

$$A = \{x \mid x \in S, P(x)\} \text{ or } A = \{x \in S \mid P(x)\}$$

means that A is the set of all elements x of S such that x satisfies the property P.

Common sets of numbers

Symbol	Meaning	Examples
$\mathbb{N}$	Natural numbers (here: positive integers)	1,2,3,…
$\mathbb{Z}$	Integers	…,-2,-1,0,1,2,…
$\mathbb{Q}$	Rational numbers: a/b, a,b ∈ $\mathbb{Z}$ and b ≠ 0	1/2, -3, 0.75
$\mathbb{R}$	Real numbers:union of the rational numbers with irrational numbers	-2, 1/2, √2, π

√2 ,and π are irrational number.

Example set-builder notation

$A = \{x \in \mathbb{Z} \mid 0 < x < 7\}$

$x \in \mathbb{Z}$: x must be an integer.

| means “such that”.

$0 < x < 7$: x must satisfy this property

Therefore, A = {1,2,3,4,5,6}.

Example set-builder notation

$B = \{x \in \mathbb{Z} \mid x > 0\}$

The domain is the integers.

The condition selects only positive values.

$B = \{1,2,3,4,\ldots\}$

Sets

A set is determined by its elements and not by any particular order in which the element might be listed.

Example, $A=\{1, 2, 3, 4\}$,

A might just as well be specified as $\{2, 3, 4, 1\}$ or $\{4, 1, 3, 2\}$

The elements making up a set are assumed to be **distinct**, we may have duplicates in our list, only one occurrence of each element is in the set.

Example

$\{a, b, c, a, c\} \longrightarrow \{a, b, c\}$

$\{1, 3, 3, 5, 1\} \longrightarrow \{1, 3, 5\}$

Sets

- Use uppercase letters *A*, *B*, *C* … to denote sets, lowercase denote the elements of set.
- The symbol ∈ stands for 'belongs to'
- The symbol ∉ stands for 'does not belong to'

Example

$X=\{ a, b, c, d, e \}$, $b \in X$ and $m \notin X$

$A=\{\{1\}, \{2\}, 3, 4\}$, $\{2\} \in A$ and $1 \notin A$

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Some Symbols Used With Set Builder Notation

The standard form of notation for this is called "set builder notation".

For instance, $\{x \mid x \text{ is an odd positive integer}\}$ represents the set $\{1, 3, 5, 7, 9, \ldots\}$

$\{x \mid x \text{ is an odd positive integer}\}$ is read as

"the set consisting of all x such that x is an odd positive integer".

The vertical bar, " $\mid$ ", stands for "such that"

Other "short-hand" notation used in working with sets

"$\forall$" stands for "for every"

"$\exists$" stands for "there exists"

"$\bigcup$" stands for "union"

"$\bigcap$" stands for "intersection"

"$\subseteq$" stands for "is a subset of"

"$\subset$" stands for "is a (proper) subset of"

"$\not\subset$" stands for "is a not a (proper) subset of"

"$\varnothing$" stands for the "empty set"

"$\in$" stands for "is an element of"

"$\notin$" stands for "is not an element of"

"$\times$" stands for "cartesian cross product"

"$=$" stands for "is equal to"

Subset

If every element of A is an element of B, we say that A is a subset of B and write $A \subseteq B$.

$A=B$, if $A \subseteq B$ and $B \subseteq A$

The empty set ($\emptyset$) is a subset of every set.

Example $A=\{1, 2, 3\}$

Subset of A,

$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

Note: A* is a subset of *A

Proper Subset

If $A \subseteq B$ and B contains an element that is not in A, then we say "A is a **proper subset** of B": $A \subset B$ or $B \supset A$.

Formally: $A \subseteq B$ means $\forall x\,[x \in A \rightarrow x \in B]$.

For all sets: $A \subseteq A$.

Note: If *A* is a subset of *B* and *A* does not equal B, we say that *A* is a proper subset of *B* (*A* ⊆ *B* and *A*≠*B* (*B* ⊈ *A*))

Example

- Let, $A=\{1, 2, 3\}$

  Proper subset of A,

  $\emptyset$, {1}, {2}, {3}, {1, 2}, {1, 3}, {2, 3}

- Let, $B=\{1, 2, 3, 4, 5, 6\}$

  A is proper subset of B.

Example

A = {a,b,c,d,e,f,g,h}

B = {b,d,e}

C = {a,b,c,d,e}

D = {r,s,d,e}

Proper subset of A ??

Empty Sets

The **empty set** $\varnothing$ or {} but not $\{\varnothing\}$
is the set without elements.

Note:

- Empty set has no elements
- Empty set is a subset of any set
- There is exactly one empty set
- Properties of empty set:
  $A \cup \varnothing = A$, $A \cap \varnothing = \varnothing$
  $A \cap A' = \varnothing$, $A \cup A' = U$
  $U' = \varnothing$, $\varnothing' = U$

Example

$\emptyset = \{x \mid x \text{ is a real number and } x^2 = -3\}$

$\emptyset = \{x \mid x \text{ is positive integer and } x^3 < 0\}$

Equal Sets

The sets A and B are **equal** (A=B) if and only if each element of A is an element of B and vice versa.

Formally: $A=B$ means $\forall x\,[x \in A \leftrightarrow x \in B]$.

Example

$A = \{a, b, c\}$, $B=\{b, c, a\}$,

$A = B$

$C = \{1, 2, 3, 4\}$

$D = \{x \mid x \text{ is a positive integer and } 2x < 10\}$,

$C = D$

Equivalent Sets

Two sets, A and B, are equivalent if there exists a one-to-one correspondence between them.

When we say sets "have the same size", we mean that they are equivalent.

Example

Set A= {A, B, C, D, E} and **Set** B={1, 2, 3, 4, 5}

Note:

- An equivalent set is simply a set with an **equal number of elements**.
- The sets do not have to have the same exact elements, just the same number of elements.

Finite Sets

A set A is finite

if it is empty

or

if there is a natural number *n*
such that set A is equivalent to
{1, 2, 3, . . . *n*}.

Example

$A = \{1, 2, 3, 4\}$

$B = \{x \mid x \text{ is an integer}, 1 \leq x \leq 4\}$

Note:

There exists a nonnegative integer n such that A has n elements (A is called a finite set with n elements)

Infinite Sets

- An infinite set is a set whose **elements can not be counted**.
- An infinite set is one that has **no last element**

A set *A* is infinite

if there is NOT a natural number *n* such that
set *A* is equivalent to {1, 2, 3, . . . *n*}.

Infinite sets are uncountable.
Are all infinite sets equivalent?

A set is infinite if it is equivalent to a proper
subset of itself!

An infinite set is a set that can be placed into a **one-to-one correspondence** with a proper subset of itself.

Example

Infinite sets

$Z = \{x |\ x \text{ is an integer}\}$

or $Z = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

$S = \{x |\ x \text{ is a real number and } 1 \leq x \leq 4\}$

$D = \{x \mid x \text{ is an integer}, \ x > 0\}$

Finite Sets

$C = \{5, 6, 7, 8, 9, 10\}$

$B = \{x \mid x \text{ is an integer}, 10 < x < 20\}$

Universal Set

- Sometimes we are dealing with sets all of which are subsets of a set U.
- This set U is called a universal set or a universe.
- The set U must be explicitly given or inferred from the context

Universal Set

Typically we consider a set A
a part of a **universal set** $\mathcal{U}$,
which consists of all possible elements.

To be entirely correct we should say

$$\forall x \in \mathcal{U}\ [x \in A \leftrightarrow x \in B]$$

instead of

$$\forall x\ [x \in A \leftrightarrow x \in B] \text{ for } A=B.$$

Note that $\{ x \mid 0<x<5\}$ is can be ambiguous.
Compare $\{ x \mid 0<x<5, x \in N \}$ with $\{ x \mid 0<x<5, x \in Q \}$

Example

- The sets $A=\{1,2,3\}$, $B=\{2,4,6,8\}$ and $C=\{5,7\}$
- One may choose $U=\{1,2,3,4,5,6,7,8\}$ as a universal set.
- Any superset of U can also be considered a universal set for these sets A, B, and C.

For example, $U=\{x \mid x \text{ is a positive integer}\}$

Cardinality of Set

- Let S be a finite set with n distinct elements, where $n \geq 0$.

- Then we write $|S|=n$ and say that the cardinality (or the number of elements) of S is n.

Example

$A = \{1, 2, 3\}$, $|A|=3$

$B = \{a,b,c,d,e,f,g\}$, $|B|=7$

Power Set

- The set of all subsets of a set A, denoted $P(A)$, is called the power set of A.

$$P(A) = \{X \mid X \subseteq A\}$$

If $|A|=n$, then $|P(A)| = 2^n$

Example $A=\{1,2,3\}$

The power set of A,

$P(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1,2\}, \{1,3\}, \{2,3\}, \{1,2,3\}\}$

Notice that $|A| = 3$, and $|P(A)| = 2^3 = 8$

Example

$A = \{SQL, Python\}$ $B = \{SQL, Python, Spark\}$

$A \subseteq B$ and $A \neq B$, so A is a proper subset of B.

$Python \in A$, but $\{Python\} \subseteq A$. Membership and subset are different relationships.

$|A| = 2$ and $P(A) = \{\emptyset, \{SQL\}, \{Python\}, \{SQL, Python\}\}$.

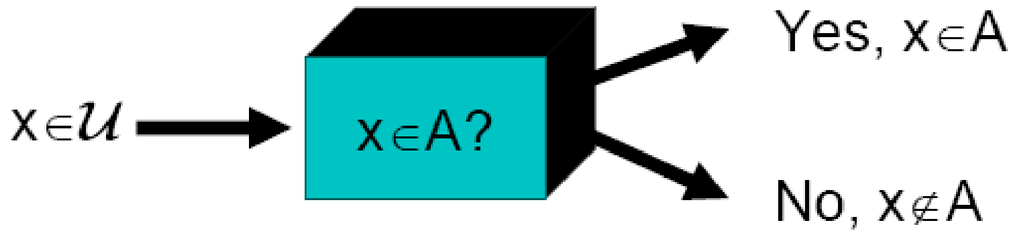
How to Think of Sets

The elements of a set do not have an ordering, hence {a,b,c} = {b,c,a}

The elements of a set do not have multitudes, hence {a,a,a} = {a,a} = {a}

All that matters is: “Is x an element of A or not?”

The size of A is thus the number of *different* elements



Thank You